

Analysis of Mass Flow and Momentum in an Axisymmetric
Air Jet Using Pitot Tube Measurements

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Abstract

This experiment investigates the flow characteristics of an axisymmetric air jet through analysis of velocity, mass flow rate, and momentum through measurements taken by a pitot tube and pressure transducer. The experiment involved analyzing the flow field at various downstream locations $\frac{x}{D} = 1, 3, 5, 7$ and incremented radii from the centerline to understand the jet's behavior. Velocity, mass flow rate, and momentum were calculated using experimental data processed in MATLAB. The results aligned with an expected decrease in center line velocity downstream, the radial expansion of the jet downstream, and the conservation of mass and momentum. Key findings include the confirmation of the turbulent nature of the jet, shown through power-law relationships relating centerline velocity and jet diameter to their respective downstream locations. Inaccuracies in the data are attributed to potential experimental setup errors, emphasizing the need for more precision in future experiments

Introduction

An axisymmetric jet is a device that creates a flow, typically of either air or water, by way of a diameter D orifice, such that the velocity and pressure profiles remain constant around a central, usually horizontal, axis. For such a jet, the fluid spreads radially outward in a cone originating at the jet exit, until diffusion and dissipation reduce the jet flow to that of ambient conditions. A typical axisymmetric jet consists of a round nozzle so that the flow distributes uniformly around the centerline. The centerline is taken from the center of the nozzle and extended in the direction of the flow. Flow radius R is taken from the centerline outward, with $R = 0$ being on the centerline. The flow velocity decreases along the centerline and radially. The mass flow rate will increase slightly due to entrainment. The flow momentum will be conserved along the centerline but decreases as diffusion takes effect (Cushman-Roisin, 1974).

A pitot tube is a flow measurement device that is typically used to measure a fluid velocity at some point in a flow field through a change in stagnation and static pressure. It is placed parallel to the direction of flow, with the opening toward the oncoming flow. Stagnation pressure is measured from the front of the pitot tube, and static pressure is measured either from a port along the side of the tube or by using ambient, no-flow conditions (Anderson & Cadou, n.d.). A port along the side of the tube can be used to measure static pressure because of the no-slip condition, which states that the velocity of a fluid on the surface of an object will be zero due to friction forces (Gerhart et al., 2019). Utilizing the Bernoulli Equation, along with the no-slip condition, the velocity of the flow can be determined. The derivation of the velocity can be found below in equation (2).

For this experiment, a pitot tube is used with an Omega PX-138 pressure transducer connected to an NI USB-6211 analog-to-digital converter. The Omega PX-138 has a pressure range of ± 0 to 0.3 psi and a linear relationship between the pressure measured and the output voltage. A calibration curve with a calibration constant $C = 823 \text{ Pa/V}$ is provided. The calibration curve can be found in Appendix A.1. The goal of the experiment is to, using a pitot tube, measure both the mass flow rate and momentum of the airflow out of a steady-state axisymmetric jet of diameter D . The jet is powered by compressed air and is controlled by a manual valve. A pitot tube will be placed in the flow field of the jet at varying distances, with the height being varied from the centerline to the approximate edge of the flow field. The distance of the pitot tube is determined by x/D , where x is the location downstream. The x/D values used in this experiment are $x/D = 1, 3, 5, 7$. The radius R will be measured in spaced increments from 0 to the edge of the flow, with values being recorded at each radius. Once all the data is taken, it is processed in MATLAB to find velocities, mass flow, and momentum.

Fundamental Equations

Gage Pressure

$$P_{\text{gage}} = C * (V_{\text{measured}} - V_{\text{offset}}) \quad (1)$$

Velocity V

From the Bernoulli Equation with the no-slip condition applied between points A and B, where B is the surface on which the no-slip condition is applied (Anderson & Cadou, n.d.),

$$P_A + \frac{1}{2}\rho V_A^2 = P_B + \frac{1}{2}\rho V_B^2$$

Or

$$P_1 + \frac{1}{2}\rho V_1^2 = P_0 + 0$$

Solving for V_1 ,

$$V_1 = \sqrt{\frac{2(P_0 - P_1)}{\rho}}$$

Taking $P_0 - P_1 = P_{\text{gage}}$,

$$V_1 = \sqrt{\frac{2 \cdot P_{\text{gage}}}{\rho}} \quad (2)$$

Mass Flow Rate general equation

$$\dot{m} = \int_A \rho_{air} V dA$$

Assume incompressible flow,

$$\dot{m} = \rho_{air} \int_A V dA$$

$$dA = 2\pi r dr$$

$$\dot{m} = 2\pi\rho_{air} \int_0^r V r dr$$

Applying a numerical integration for the integral by trapezoidal summation,

$$\dot{m} = 2 \cdot \pi \cdot \rho_{air} \cdot \sum_{i=1}^n \frac{V(r_{i-1}) \cdot r_{i-1} + V(r_i) \cdot r_i}{2} \cdot \Delta r_i \quad (3)$$

Momentum general equation

$$J = \int_A \rho_{air} V^2 dA$$

Assume incompressible flow,

$$J = \rho_{air} \int_A V^2 dA$$

$$dA = 2\pi r dr$$

$$J = 2\pi\rho_{air} \int_0^r V^2 r dr$$

Applying a numerical integration for the integral by trapezoidal summation,

$$J = 2 \cdot \pi \cdot \rho_{air} \cdot \sum_{i=1}^n \frac{V^2(r_{i-1}) \cdot r_{i-1} + V^2(r_i) \cdot r_i}{2} \cdot \Delta r_i \quad (4)$$

$$\Delta r_i = r_i - r_{i-1} = n_{radial\ positions} \quad (5)$$

Variables

$C = \text{Omega PX} - 138 \text{ Experimental Calibration Constant} = 823 \text{ Pa/V}$

$Q_{measured} = \text{Measured Flow Rate} = 62 \text{ SCFM}$

$P_{ac} = \text{Measured Flow Pressure} = 15 \text{ psig}$

$\rho_{air} = \text{Density of Air} = 1.225 \text{ kg/m}^3$

$D = \text{Diameter of Nozzle}$

$r = \text{Radius of the Jet}$

$U(r) = \text{Velocity at Radius } r$

$U_{CL} = \text{Centerline Velocity}$

$x = \text{Horizontal Distance}$

$x_{origin} = \text{Left most point on the traverse} = 0 \text{ mm}$

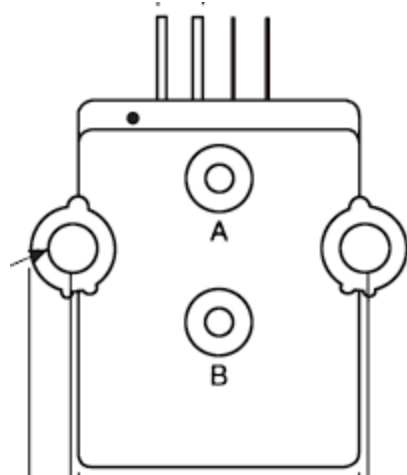
$x_o = \text{Reference location at the nozzle for horizontal movement} = 220 \text{ mm}$

$z = \text{Vertical distance}$

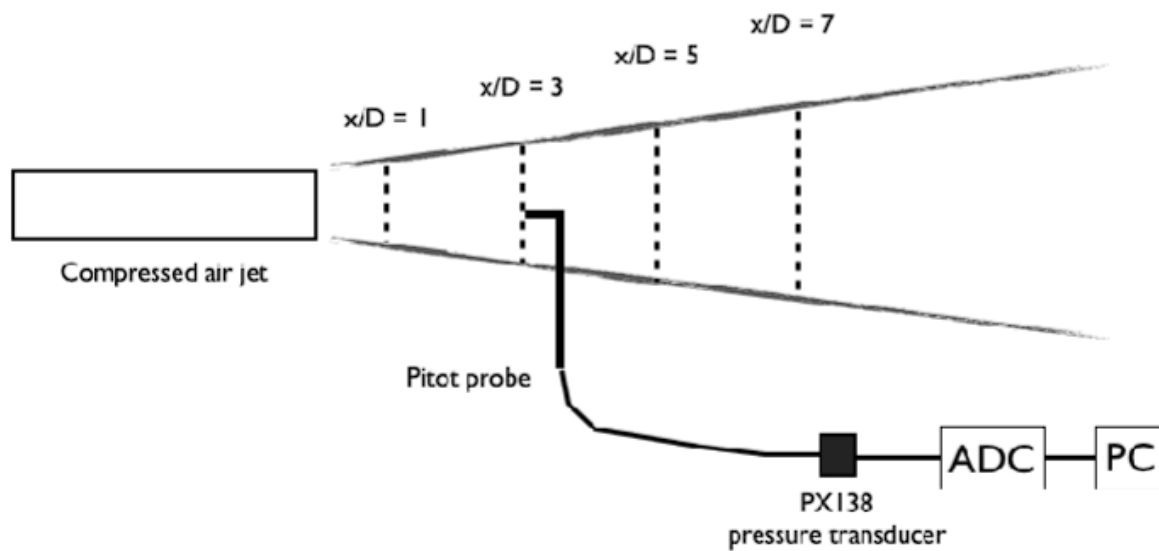
$z_o = \text{Reference location at the centerline for vertical movement} = 50 \text{ mm}$

Experimental Setup

This experiment analyzes an axisymmetric air jet using a Pitot tube connected to an Omega PX-138 pressure transducer. The transducer is powered by an 8 VDC supply from a BK Precision 1760A power source, which converts the pressure into a voltage output. The voltage output is then recorded and processed in MATLAB to determine the corresponding velocity values. The transducer's positive pressure port (port B) connects to the stagnation pressure port of the Pitot tube. Port A is left open to the ambient air to measure atmospheric pressure. The output range of pin 3 of the Omega PX-138 is 3.0 VDC to 6.0 VDC. Pin 3 is connected to a low pass filter, which connects to the positive side of AI0. Also connected to the output of the low pass filter is a Siglent SDS1102CML oscilloscope for real-time output voltage monitoring. Pin 2 of the Omega PX-138 is connected to the negative side of AI0 and the ground of the power supply. See Appendix B.1 for a more detailed description of the circuit, as well as a picture from this experiment.



(Figure 1) Omega PX-138 Pressure Transducer, Pinout from left to right:
 V_{in} , GND , V_{out} and signal, no connection



(Figure 2) High-level overview of axisymmetric jet and pitot probe setup. See Appendix B.2 for pictures of the experimental setup and a more detailed explanation of adjustments that can be made.

Now that the circuit has been set up, the pitot probe can be properly positioned for data acquisition. The pitot probe inlet should be as close to the center of the nozzle as possible. Going off of eyesight will be accurate enough. The pitot probe should be placed close to the nozzle. Turn on the power supply and oscilloscope and take note of the mean reading. This value is the offset voltage when there is no flow. The expected offset value is around 3.5 VDC. Now, turn on the flow and take note of the new reading on the oscilloscope. There should be a noticeable increase in the voltage compared to

no flow. Take note of this voltage. Using Equation (1) and calibration constant C , calculate the gage pressure. Then, using Equation (2) and ρ_{air} , calculate the velocity of the flow. If the value calculated seems reasonable, trials at varying x/D and R can be recorded for analysis.

See Appendix C.1 for the data acquisition script that was used in this experiment. There are two variables whose values need to be chosen, which are the sampling rate and time. The cutoff frequency of the low pass filter is 50 Hz, so by the Nyquist-Shannon Sampling theorem, the sampling rate needs to be at least 100 Hz. For this experiment, a sampling rate of 250 Hz was chosen and was implemented in line 22 of the data acquisition script. The sampling time needs to be sufficiently long so that any disturbances that may occur do not drastically affect the whole dataset. For this experiment, a sampling time of 25 seconds was chosen and was implemented in line 25. Once these two variables have been implemented into the code, make sure the jet is off, and run the code, making sure to adjust the names of the files being saved in such a way that the data they contain can be easily determined. For the first dataset, a name such as 'Offset' will suffice. Be sure to update the names every time you run the code, or the data will not be stored. After the code has run, look at the generated plot and make sure it looks correct. Now that the offset voltage has been recorded and stored, the pitot probe can be moved to a variety of locations, with the script run at each location.

There will be four horizontal locations from which data will be acquired; $x/D = 1, 3, 5$, and 7 . For each location, the process of recording data will be the same, however for $x/D = 5$ and 7 , there will be more radial positions required to represent the expansion of the airflow more accurately. It is recommended that for the first two, 10 radial positions are used, and for the third, 15, and fourth, 20 locations. For the first measurement for each x/D location, move the pitot probe to the edge of the jet using the vertical traverse and watch the oscilloscope for a change in voltage back to the offset, signaling the probe has left the flowfield. The probe should be moved just below this point so that it is on the edge, but still in the flow. Take note of the centerline value of the traverse, as well as the jet radius value. From the difference between these two and the recommended number of radii, an increment for R can be determined. Check the output data and plot and make sure everything looks correct. This process can be repeated for each x/D location, at each radial position. Once all data has been acquired, data analysis can be performed. Before ending the experiment, the flow rate and pressure should be recorded from the dial indicators. The jet can then be turned off. For this experiment, $x = 0$, or the reference for horizontal movement of the probe, is at the furthest point away on the traverse from the nozzle. x_0 is up against the nozzle. To calculate the locations x of $\frac{x}{D}$, the diameter D

is multiplied by the desired ratio (1, 3, 5, 7), and subtracted from x_0 . For this experiment, this resulted in x locations 194.5, 140.5, 181.5, and 22.5 mm from the origin. For the z axis, zero is taken from the bottom-most point of the traverse. z_0 , the reference point for the z axis is taken from the centerline. For this experiment, $z_0 = 50$ mm from the origin.

Results

Velocity Profile Analysis

The velocity profiles at each x/D location (1, 3, 5, 7) are obtained by plotting $\frac{U(r)}{U_{CL}}$ vs $\frac{r}{D}$, U_{CL} is the centerline velocity as shown in Appendix D.1. The velocity $U(r)$ is calculated using equations (1) and (2) at each point, with the pressures in equation (1) being the average of the locations dataset. The calculated profiles are what is expected for an axisymmetric jet.

Centerline Velocity U_{CL} vs $\frac{x}{D}$

The centerline velocity U_{CL} is plotted against $\frac{x}{D}$ to show the decrease in velocity as the distance from the jet decreases. This is shown in Appendix D.6. Using the standard deviation of each dataset, the standard error was calculated using,

$$S_{\bar{x}} = \frac{s}{\sqrt{N}} \quad (5)$$

The error was then plotted in the form of error bars as shown in Appendix D.2 through D.5. The 99.7% confidence interval was also calculated using the equation,

$$99.7\% \text{ confidence} = \bar{x} \pm 3S_{\bar{x}} \quad (6)$$

Tabulated results,

x/D	UCL	Standard Error	CI Lower	CI Upper
1	25.987	0.00018597	25.987	25.986
3	25.099	7.5163e-05	25.099	25.098
5	22.232	0.00018078	22.233	22.232
7	18.581	0.00020745	18.582	18.581

Table (1) 99.7% confidence intervals for each U_{CL} with upper and lower bounds

Power Law Relationship Between U_{CL} and $(\frac{x}{D})^{-1}$

The centerline velocity is plotted against $(\frac{x}{D})^{-1}$ to show that the velocity follows a power law equation as the distance from the nozzle decreases, in this case to $\frac{x}{D} = 1$. A power law fit is then applied to the data, using the equation,

$$y = ax^b + c \quad (7)$$

The resulting coefficients are as follows,

General model Power2:

fitresult(x) = a*x^b+c

Coefficients (with 95% confidence bounds):

a = -0.1353 (-1.094, 0.8239)

b = -2.076 (-5.586, 1.434)

c = 26.21 (22.1, 30.32)

The plotted data and fit curve can be found in Appendix D.7. See Appendix C.2 for the relevant MATLAB script.

Jet Width $2 \cdot R_{max}$ vs $\frac{x}{D}$

The jet width, defined as twice the radius containing most of the jet's velocity, was analyzed for each $\frac{x}{D}$ and plotted against $\frac{x}{D}$. A power law fit is then applied to the data, using the equation,

$$y = ax^b \quad (8)$$

The resulting coefficients are as follows,

General model Power1:

fitresult(x) = a*x^b

Coefficients (with 95% confidence bounds):

a = 0.04343 (0.03777, 0.04908)

b = 0.3028 (0.2193, 0.3863)

The plotted data and fit curve can be found in Appendix D.8. See Appendix C.2 for the relevant MATLAB script.

Mass Flow Rate ($\dot{m}$) vs $\frac{x}{D}$

The mass flow rates were calculated from the velocity profiles at each $\frac{x}{D}$ location using Equation (3) and Equation (5). Utilizing the flow rate meter reading of 62 SCFM, the theoretical max mass flow rate of the jet was calculated at $0.0352778 \frac{kg}{s}$. Prior to finishing all the data analysis for the mass flow rate, the calculated flow rate at $\frac{x}{D} = 1$ was compared to the theoretical as a sanity check. See Appendix D.9 for the plotted data. See Appendix C.2 for the relevant MATLAB script.

Momentum (J) vs $\frac{x}{D}$

The momentum J The jet was calculated at each downstream location by integrating the momentum distribution across the jet profile using equation (4). See Appendix D.10 for the plotted data. See Appendix C.2 for the relevant MATLAB script.

Discussion of Results

Based on the data gathered and analyzed, the asymmetric jet exhibits turbulent behavior, which is expected from this kind of jet. This is supported by decay that is seen in the power laws in Appendix D.7 and D.8. For D.7, as the relationship $(\frac{x}{D})^{-1}$ approaches 1, the centerline velocity approaches that of the jet at the nozzle, shown by the asymptotic behavior at $(\frac{x}{D})^{-1} = 1$. Similarly, as $(\frac{x}{D})^{-1}$ approaches 0, the centerline velocity approaches that of the ambient air, which is 0. More locations further away from the jet would be needed to accurately show this, or perhaps a different version of the power law would give a better understanding of the relationship. For D.8, the relationship between the jet's diameter and the location $\frac{x}{D}$ shows that as the flow moves further away from the nozzle, the jet expands. From analysis, the first three $\frac{x}{D}$ locations display linear behavior, with only the fourth location beginning to deviate. This is the result of entrainment and the turbulent nature of the flow. As the flow goes downstream, surrounding ambient air is pulled into the flow, causing vortices and eddies to appear, resulting in a sharp decrease in velocity as the diameter expands.

The results also suggest that the conservation of mass and momentum is satisfied for the jet as $\frac{x}{D}$ increases. The mass flow rate increases with distance from the orifice, which is expected for a turbulent jet because of entrainment. As the jet spreads radially, it entrains surrounding air, increasing the total mass flow rate without

additional mass being injected from the source. This entrainment allows the jet to expand while still conserving the total momentum along the centerline. Appendix D.9 and D.10 both support this conclusion. The relationship shown for both mass and momentum is near constant across all $\frac{x}{D}$ locations (Labus & Symons, 1972).

There are some sources of error that may have resulted in some values deviating from what was expected. The first major source of error that could have played a large role in this experiment is that the pitot probe was not centered or leveled enough. If it was not centered, then it would not be lying on the centerline, resulting in inaccurate calculations for the velocities. If the probe was not level, then the voltage readings due to changes in pressure would not be accurate, and would also throw off the velocity calculations. A second major source of error could be that the vertical measurement of the edge of the jet was not correct. Perhaps the probe was placed too high, outside of the flow, or too low, not on the edge of the flow. Being too low would explain the deviation of Appendix D.8.

Conclusions

This experiment successfully characterized and displayed the behavior of an axisymmetric turbulent jet, affirming expected results for velocity decrease, jet expansion, mass flow rate, and momentum conservation. The measured velocity along the center line decreased with increasing $\frac{x}{D}$, which is a key characteristic of turbulent flow with entrainment. The diameter of the jet increased linearly for the first three $\frac{x}{D}$ locations but deviated on the fourth. This may be the result of incorrect or inaccurate experimental setup and should be a focus for future experiments to ensure accuracy. The mass flow rate and momentum were both conserved throughout the flow along the centerline, as shown in Appendix D.9 and D.10, with their near-constant value across all $\frac{x}{D}$ locations.

This experiment was not perfect. Much of the setup was left to human accuracy, which is good enough for an instructional lab, but should be minimized as much as possible were these results to be published. The key areas for improvement of the setup would involve calculating all of the pitot probe dimensional parameters, instead of eyeballing. This would result in a more accurate measurement of flow pressure, and in turn, a more accurate measurement of velocity. Another area that could use improvement is the initial max radial positioning of the pitot probe in the flow. If this is not correctly set up, the data could be inconsistent across $\frac{x}{D}$ at best, or missing fundamental data at worst. The number of radial positions chosen for each $\frac{x}{D}$ location worked well and allowed a clear picture of the flow to be realized. Choosing more

positions will always be better, but there may not be a reason beyond consistency in this case.

References

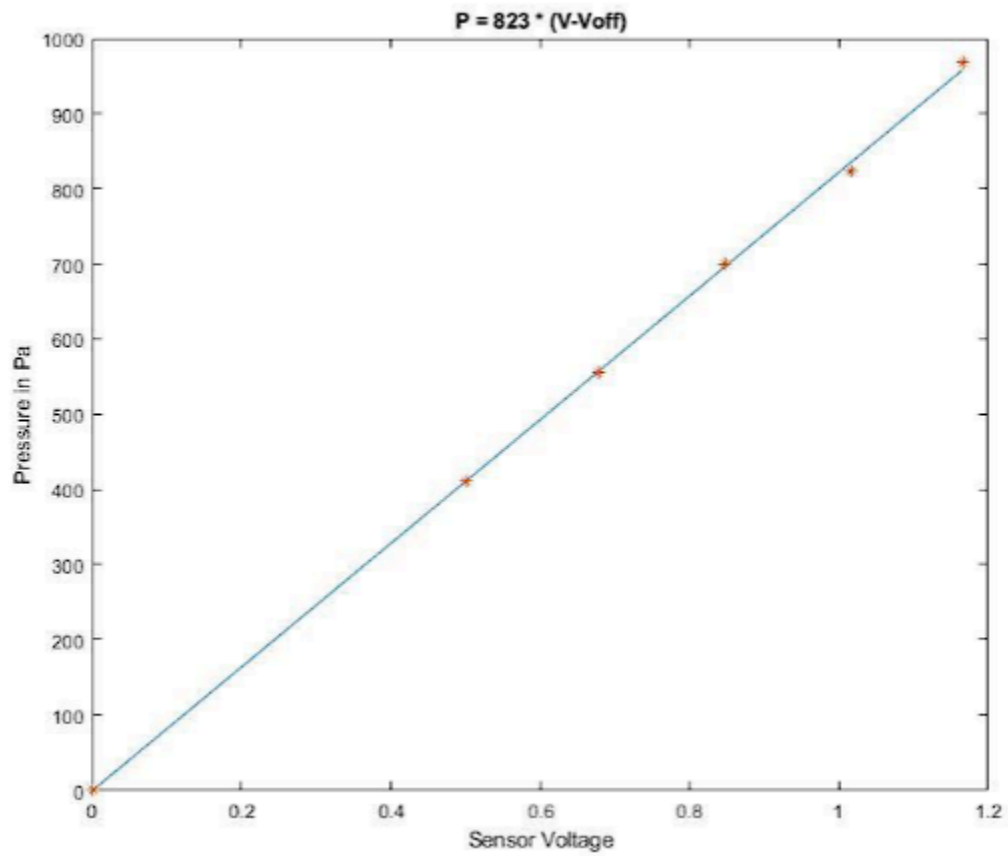
Anderson, J. D., & Cadou, C. (n.d.). Pitot Tube: Measurement of Airspeed [PDF]. In *Fundamentals of Aerodynamics* (p. 230).

Cushman-Roisin, B. (1974). Turbulent jets [PDF]. In *ENVIRONMENTAL FLUID MECHANICS* (pp. 153–159). John Wiley & Sons.
<https://cushman.host.dartmouth.edu/books/EFM/chap9.pdf>

Gerhart, P. M., Gerhart, A. L., & Hochstein, J. I. (2019). *Munson, Young and Okiishi's Fundamentals of Fluid Mechanics, 8th edition EMEA Edition*.

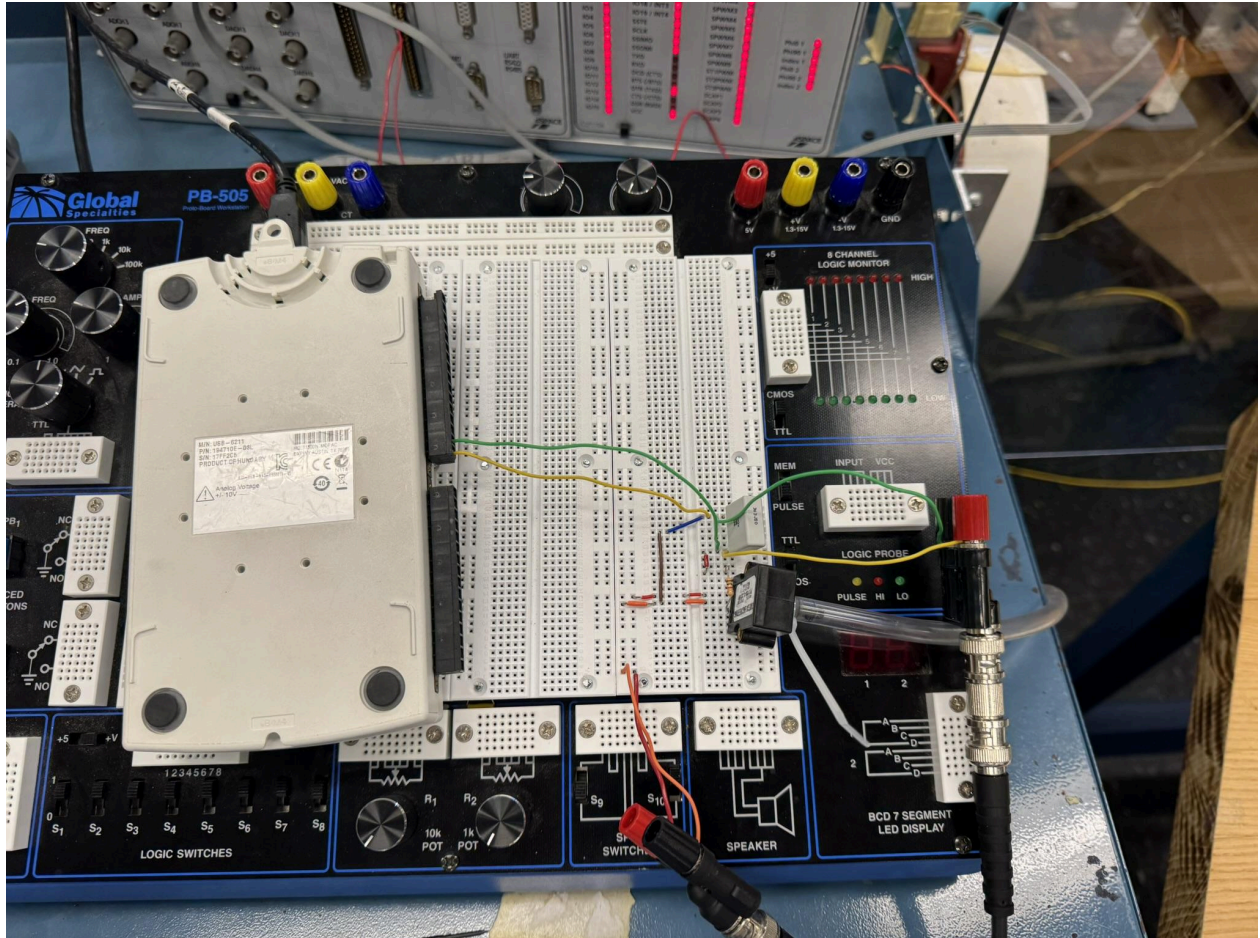
Labus, T. L., & Symons, E. P. (1972). *Experimental Investigation of an Axisymmetric Free Jet with an Initially Uniform Velocity Profile*.

Appendix A - Given Figures

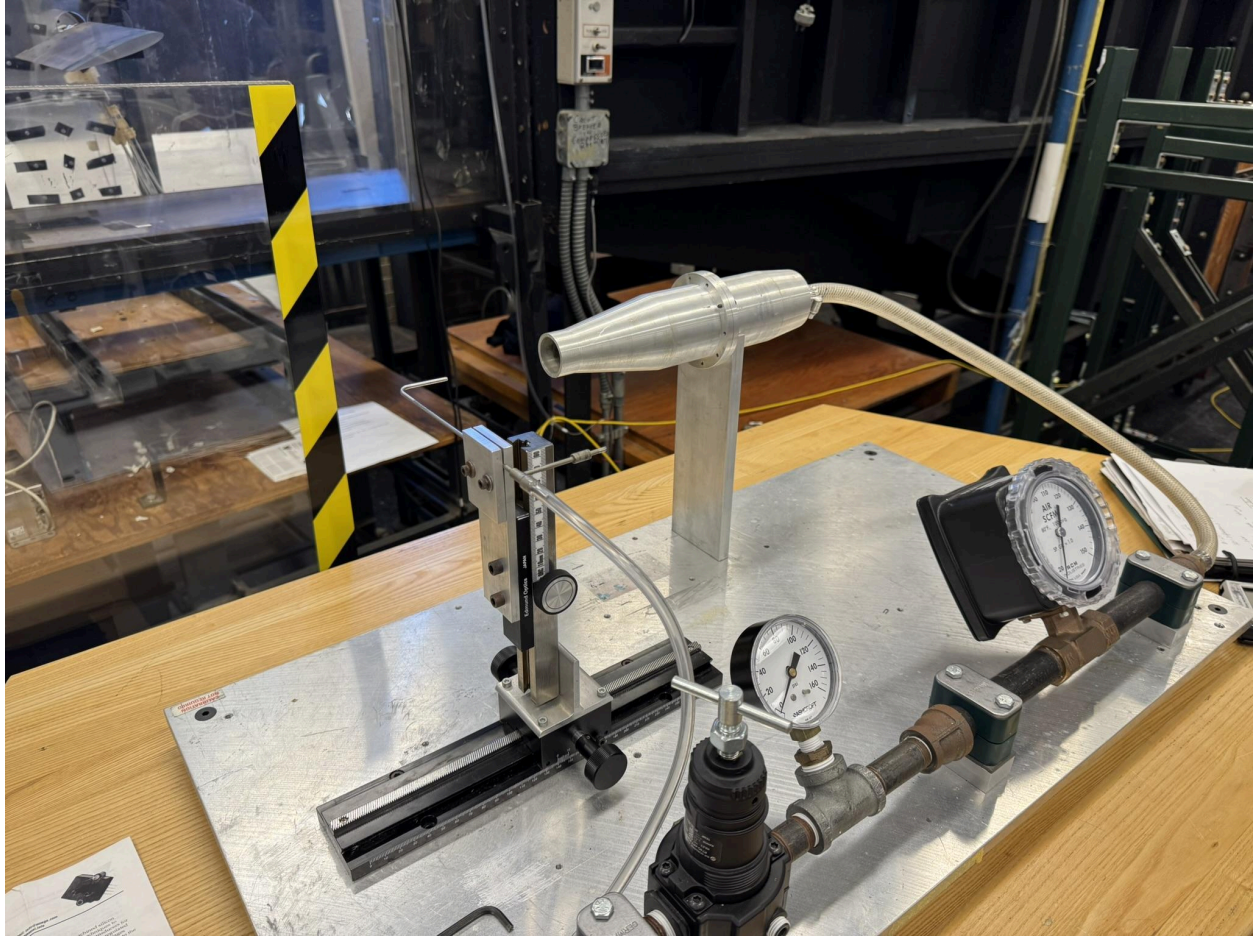


(1) The calibration curve used to calculate P_{gage} , using $C = 823 \text{ Pa/V}$

Appendix B - Pictures



- (1) Circuit setup connecting the Omega PX-138 to the NI USB-6211 via an RC circuit. 8 VDC is input on the bottom. The positive side is then connected to pin 1, which is the topmost pin pictured. Pin 2 is connected to ground. Pin 3, which is the bottommost pin used here, is connected to a resistor, and then a capacitor. The output of the capacitor is connected to ground. The oscilloscope is connected across the capacitor, as shown on the right. The NI USB-6211 is also connected across the capacitor, as shown on the left.



- (2) Experimental setup of the axisymmetric jet and pitot probe. The jet is at a fixed location. The pitot probe is on a traverse that can move in the x and z directions via knobs and distance markers. Control in the y direction is adjusted manually using a screw clamp at the top. Rotation can also be adjusted this way.

Appendix C - MATLAB Script

```

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% MMAE 315, IIT, Fall 2024
% A program for reading data with MATLAB and an NI DAQ board, to be used in
% Laboratory 2
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

clear;
clc;

daq.getDevices
daq.reset
s=daq.createSession('ni');

ch1 = addAnalogInputChannel(s, 'Dev1', 0, 'Voltage');

set(ch1, 'Range', [-5.0 5.0]);

%Sampling Frequency [Hz]
s.Rate = 250;

%Sampling duration [s]
s.DurationInSeconds = 25;

set(ch1, 'TerminalConfig', 'Differential');

[XD7_20, t] = startForeground(s);

%Desired plot
plot(t, XD7_20)

save('XD7_20.mat', 'XD7_20')
load('XD7_20.mat')
Mean = mean(XD7_20)
Standard_deviation = std(XD7_20)

```

- (1) Data acquisition script used in this experiment. Sampling Frequency and Sampling Duration are to be set before the experiment begins and should remain constant throughout the experiment to ensure consistency. In this script, the current x/D and R are 7 and 20. This is reflected in the filename XD7_20. All instances where such a filename occurs should be changed to reflect the current data being taken, and should contain enough information for easy determination of what the file contains. If the filenames are not changed between runs, the data will be overwritten, and reruns will be needed.

```

% xd1_mean(1) = mean(XD1_1);
% xd1_mean(2) = mean(XD1_2);
% xd1_mean(3) = mean(XD1_3);
% xd1_mean(4) = mean(XD1_4);
% xd1_mean(5) = mean(XD1_5);
% xd1_mean(6) = mean(XD1_6);
% xd1_mean(7) = mean(XD1_7);
% xd1_mean(8) = mean(XD1_8);
% xd1_mean(9) = mean(XD1_9);
% xd1_mean(10) = mean(XD1_10);
%
% xd3_mean(1) = mean(XD3_1);
% xd3_mean(2) = mean(XD3_2);
% xd3_mean(3) = mean(XD3_3);
% xd3_mean(4) = mean(XD3_4);
% xd3_mean(5) = mean(XD3_5);
% xd3_mean(6) = mean(XD3_6);
% xd3_mean(7) = mean(XD3_7);
% xd3_mean(8) = mean(XD3_8);
% xd3_mean(9) = mean(XD3_9);
% xd3_mean(10) = mean(XD3_10);
%
% xd5_mean(1) = mean(XD5_1);
% xd5_mean(2) = mean(XD5_2);
% xd5_mean(3) = mean(XD5_3);
% xd5_mean(4) = mean(XD5_4);
% xd5_mean(5) = mean(XD5_5);
% xd5_mean(6) = mean(XD5_6);
% xd5_mean(7) = mean(XD5_7);
% xd5_mean(8) = mean(XD5_8);
% xd5_mean(9) = mean(XD5_9);
% xd5_mean(10) = mean(XD5_10);
% xd5_mean(11) = mean(XD5_11);
% xd5_mean(12) = mean(XD5_12);
% xd5_mean(13) = mean(XD5_13);
% xd5_mean(14) = mean(XD5_14);
% xd5_mean(15) = mean(XD5_15);
%
% xd7_mean(1) = mean(XD7_1);
% xd7_mean(2) = mean(XD7_2);
% xd7_mean(3) = mean(XD7_3);
% xd7_mean(4) = mean(XD7_4);
% xd7_mean(5) = mean(XD7_5);
% xd7_mean(6) = mean(XD7_6);
% xd7_mean(7) = mean(XD7_7);
% xd7_mean(8) = mean(XD7_8);
% xd7_mean(9) = mean(XD7_9);
% xd7_mean(10) = mean(XD7_10);
% xd7_mean(11) = mean(XD7_11);
% xd7_mean(12) = mean(XD7_12);
% xd7_mean(13) = mean(XD7_13);
% xd7_mean(14) = mean(XD7_14);
% xd7_mean(15) = mean(XD7_15);
% xd7_mean(16) = mean(XD7_16);
% xd7_mean(17) = mean(XD7_17);
% xd7_mean(18) = mean(XD7_18);
% xd7_mean(19) = mean(XD7_19);
% xd7_mean(20) = mean(XD7_20);

load('xd1_mean.mat')
load('xd3_mean.mat')
load('xd5_mean.mat')
load('xd7_mean.mat')
load('xd1_std.mat')
load('xd3_std.mat')
load('xd5_std.mat')
load('xd7_std.mat')
load('Offset_Part6.mat')

offset = mean(Offset_Part6);

```

```

C = 823; % Pa/V
rho = 1.225; % kg/m^3
jet_diameter = 29.5 ./ 1000; % m

xd1_velocity = zeros(1, length(xd1_mean));
xd3_velocity = zeros(1, length(xd3_mean));
xd5_velocity = zeros(1, length(xd5_mean));
xd7_velocity = zeros(1, length(xd7_mean));

for i = 1 : length(xd1_mean)

    pgage = C * abs(xd1_mean(i) - offset);

    xd1_velocity(i) = sqrt((2 * pgage) / rho);

end

for i = 1 : length(xd3_mean)

    pgage = C * abs(xd3_mean(i) - offset);

    xd3_velocity(i) = sqrt((2 * pgage) / rho);

end

for i = 1 : length(xd5_mean)

    pgage = C * (xd5_mean(i) - offset);

    xd5_velocity(i) = sqrt((2 * pgage) / rho);

end

for i = 1 : length(xd7_mean)

    pgage = C * abs(xd7_mean(i) - offset);

    xd7_velocity(i) = sqrt((2 * pgage) / rho);

end

x1_radial_velocity = xd1_velocity ./ xd1_velocity(10);
x3_radial_velocity = xd3_velocity ./ xd3_velocity(10);
x5_radial_velocity = xd5_velocity ./ xd5_velocity(15);
x7_radial_velocity = xd7_velocity ./ xd7_velocity(20);

xd1_radius = [22 18 16 14 12 10 7 5 3 0] ./ 1000; % m
xd3_radius = [29.5 26 23 20 17 14 11 8 5 0] ./ 1000; % m
xd5_radius = [36 34 32 30 28 26 24 22 20 18 15 12 8 4 0] ./ 1000; % m
xd7_radius = [39 37 35 33 31 29 27 25 23 21 19 17 15 13 11 9 7 5 3 0] ./ 1000; % m

x1_rd = xd1_radius ./ jet_diameter;
x3_rd = xd3_radius ./ jet_diameter;
x5_rd = xd5_radius ./ jet_diameter;
x7_rd = xd7_radius ./ jet_diameter;

figure(1)
plot(x1_rd, x1_radial_velocity, 'LineWidth', 1)
hold on
plot(x3_rd, x3_radial_velocity, 'LineWidth', 1)
plot(x5_rd, x5_radial_velocity, 'LineWidth', 1)
plot(x7_rd, x7_radial_velocity, 'LineWidth', 1)
title('U(r)/Ucl vs r/D at x/D = 1,3,5,7')
xlabel('r/D')
ylabel('U(r)/Ucl')
legend('x/D = 1', 'x/D = 3', 'x/D = 5', 'x/D = 7')

```

```

N = 7500;
xd = [1 3 5 7];

x1_stderr = xd1_std ./ sqrt(N);
x3_stderr = xd3_std ./ sqrt(N);
x5_stderr = xd5_std ./ sqrt(N);
x7_stderr = xd7_std ./ sqrt(N);

Ucl = [xd1_velocity(10) xd3_velocity(10) xd5_velocity(15) xd7_velocity(20)];
Ucl_stderr = [x1_stderr(10) x3_stderr(10) x5_stderr(15) x7_stderr(20)];

figure(6)
errorbar(1, Ucl(1), x1_stderr(10), 'o')
title('Ucl vs x/D for x/D = 1 with error')
xlabel('x/D')
ylabel('Ucl with error')

figure(7)
errorbar(3, Ucl(2), x3_stderr(10), 'o')
title('Ucl vs x/D for x/D = 3 with error')
xlabel('x/D')
ylabel('Ucl with error')

figure(8)
errorbar(5, Ucl(3), x5_stderr(15), 'o')
title('Ucl vs x/D for x/D = 5 with error')
xlabel('x/D')
ylabel('Ucl with error')

figure(9)
errorbar(7, Ucl(4), x7_stderr(20), 'o')
title('Ucl vs x/D for x/D = 7 with error')
xlabel('x/D')
ylabel('Ucl with error')

figure(20)
errorbar(1, Ucl(1), x1_stderr(10), '.')
hold on
errorbar(3, Ucl(2), x3_stderr(10), '.')
errorbar(5, Ucl(3), x5_stderr(15), '.')
errorbar(7, Ucl(4), x7_stderr(20), '.')
title('Ucl vs x/D with error')
xlabel('x/D')
ylabel('Ucl with error')
xlim([0 8])
ylim([18 27])

Ucl_err_upper = Ucl + 3 * Ucl_stderr;
Ucl_err_lower = Ucl - 3 * Ucl_stderr;

confidencetable = table(xd, Ucl, Ucl_stderr, Ucl_err_upper, Ucl_err_lower, 'VariableNames', ...
    {'x/D', 'UCL', 'Standard_Error', 'CI_Lower', 'CI_Upper'});
disp(confidencetable)

```

```

xd_inverse = xd.^-1;

%% Fit: 'Ucl vs x/D^-1'.
[xData, yData] = prepareCurveData(xd_inverse, Ucl );

% Set up fitype and options.
ft = fitype( 'power2' );
opts = fitoptions( 'Method', 'NonlinearLeastSquares' );
opts.Display = 'Off';
opts.StartPoint = [34.8065136190134 0.353452142511049 -0.92915571367613];

% Fit model to data.
[fitresult, gof] = fit( xData, yData, ft, opts );

% Plot fit with data.
figure( 'Name', 'Ucl vs x/D^-1' );
h = plot( fitresult, xData, yData );
title('Centerline Velocity vs (x/D)^-1')
legend( h, 'Ucl vs. xd_inverse', 'Ucl vs x/D^-1', 'Location', ...
        'SouthEast', 'Interpreter', 'none' );
% Label axes
xlabel( '(x/D)^-1', 'Interpreter', 'none' );
ylabel( 'Ucl', 'Interpreter', 'none' );
grid on

disp(fitresult)

```

4

```

x1_jet_diameters = xd1_radius .* 2;
x3_jet_diameters = xd3_radius .* 2;
x5_jet_diameters = xd5_radius .* 2;
x7_jet_diameters = xd7_radius .* 2;

x1_max_diameter = x1_jet_diameters(1);
x3_max_diameter = x3_jet_diameters(1);
x5_max_diameter = x5_jet_diameters(1);
x7_max_diameter = x7_jet_diameters(1);

Ucl_diameters = [x1_max_diameter x3_max_diameter x5_max_diameter x7_max_diameter];

%% Fit: 'untitled fit 1'.
[xData, yData] = prepareCurveData( xd, Ucl_diameters );

% Set up fitype and options.
ft = fitype( 'power1' );
opts = fitoptions( 'Method', 'NonlinearLeastSquares' );
opts.Display = 'Off';
opts.StartPoint = [44.1508496465753 0.289075397492033];

% Fit model to data.
[fitresult2, gof] = fit( xData, yData, ft, opts );

% Plot fit with data.
figure( 'Name', 'untitled fit 1' );
h = plot( fitresult2, xData, yData );
title('Diameter of Axisymmetric Jet vs x/D')

legend( h, 'Ucl_diameters vs. xd', 'Power Fit', 'Location', ...
        'SouthEast', 'Interpreter', 'none' );
% Label axes
xlabel( 'x/D', 'Interpreter', 'none' );
ylabel( 'Jet Diameter', 'Interpreter', 'none' );
grid on
xlim([0 8])

disp(fitresult2)

```

5


```

% Mass flow rate calculation using discrete integration (trapezoidal method)
xd1_mass_flow_rate = 0; % Initialize mass flow rate
for i = 1:length(xd1_velocity)-1
    % Average radius and velocity between points for trapezoidal integration
    r_avg = (xd1_radius(i) + xd1_radius(i+1)) / 2; % Average radius between points
    v_avg = (xd1_velocity(i) + xd1_velocity(i+1)) / 2; % Average velocity between points
    dA = 2 * pi * r_avg * (xd1_radius(i) - xd1_radius(i+1)); % Annular area segment

    % Incremental mass flow rate for the segment
    xd1_mass_flow_rate = xd1_mass_flow_rate + rho * v_avg * dA;
end

% Mass flow rate calculation using discrete integration (trapezoidal method)
xd3_mass_flow_rate = 0; % Initialize mass flow rate
for i = 1:length(xd3_velocity)-1
    % Average radius and velocity between points for trapezoidal integration
    r_avg = (xd3_radius(i) + xd3_radius(i+1)) / 2; % Average radius between points
    v_avg = (xd3_velocity(i) + xd3_velocity(i+1)) / 2; % Average velocity between points
    dA = 2 * pi * r_avg * (xd3_radius(i) - xd3_radius(i+1)); % Annular area segment

    % Incremental mass flow rate for the segment
    xd3_mass_flow_rate = xd3_mass_flow_rate + rho * v_avg * dA;
end

% Mass flow rate calculation using discrete integration (trapezoidal method)
xd5_mass_flow_rate = 0; % Initialize mass flow rate
for i = 1:length(xd5_velocity)-1
    % Average radius and velocity between points for trapezoidal integration
    r_avg = (xd5_radius(i) + xd5_radius(i+1)) / 2; % Average radius between points
    v_avg = (xd5_velocity(i) + xd5_velocity(i+1)) / 2; % Average velocity between points
    dA = 2 * pi * r_avg * (xd5_radius(i) - xd5_radius(i+1)); % Annular area segment

    % Incremental mass flow rate for the segment
    xd5_mass_flow_rate = xd5_mass_flow_rate + rho * v_avg * dA;
end

% Mass flow rate calculation using discrete integration (trapezoidal method)
xd7_mass_flow_rate = 0; % Initialize mass flow rate
for i = 1:length(xd7_velocity)-1
    % Average radius and velocity between points for trapezoidal integration
    r_avg = (xd7_radius(i) + xd7_radius(i+1)) / 2; % Average radius between points
    v_avg = (xd7_velocity(i) + xd7_velocity(i+1)) / 2; % Average velocity between points
    dA = 2 * pi * r_avg * (xd7_radius(i) - xd7_radius(i+1)); % Annular area segment

    % Incremental mass flow rate for the segment
    xd7_mass_flow_rate = xd7_mass_flow_rate + rho * v_avg * dA;
end

mass_flow_rates = [xd1_mass_flow_rate xd3_mass_flow_rate xd5_mass_flow_rate xd7_mass_flow_rate];

figure(12)
plot(xd, mass_flow_rates, 'o', 'LineWidth', 1)
title('Mass Flow vs x/D')
xlabel('x/D')
ylabel('Mass Flowrate (kg/s)')
xlim([0 8])
ylim([-0.5 0.5])
grid on

```

```

% Momentum flux calculation using discrete integration (trapezoidal method)
xd1_momentum_flux = 0; % Initialize momentum flux
for i = 1:length(xd1_velocity)-1
    % Average radius and velocity squared between points for trapezoidal integration
    r_avg = (xd1_radius(i) + xd1_radius(i+1)) / 2; % Average radius between points
    v_sq_avg = ((xd1_velocity(i)^2) + (xd1_velocity(i+1)^2)) / 2; % Average of velocity squared
    dA = 2 * pi * r_avg * (xd1_radius(i) - xd1_radius(i+1)); % Annular area segment

    % Incremental momentum flux for the segment
    xd1_momentum_flux = xd1_momentum_flux + rho * v_sq_avg * dA;
end

% Momentum flux calculation using discrete integration (trapezoidal method)
xd3_momentum_flux = 0; % Initialize momentum flux
for i = 1:length(xd3_velocity)-1
    % Average radius and velocity squared between points for trapezoidal integration
    r_avg = (xd3_radius(i) + xd3_radius(i+1)) / 2; % Average radius between points
    v_sq_avg = ((xd3_velocity(i)^2) + (xd3_velocity(i+1)^2)) / 2; % Average of velocity squared
    dA = 2 * pi * r_avg * (xd3_radius(i) - xd3_radius(i+1)); % Annular area segment

    % Incremental momentum flux for the segment
    xd3_momentum_flux = xd3_momentum_flux + rho * v_sq_avg * dA;
end

% Momentum flux calculation using discrete integration (trapezoidal method)
xd5_momentum_flux = 0; % Initialize momentum flux
for i = 1:length(xd5_velocity)-1
    % Average radius and velocity squared between points for trapezoidal integration
    r_avg = (xd5_radius(i) + xd5_radius(i+1)) / 2; % Average radius between points
    v_sq_avg = ((xd5_velocity(i)^2) + (xd5_velocity(i+1)^2)) / 2; % Average of velocity squared
    dA = 2 * pi * r_avg * (xd5_radius(i) - xd5_radius(i+1)); % Annular area segment

    % Incremental momentum flux for the segment
    xd5_momentum_flux = xd5_momentum_flux + rho * v_sq_avg * dA;
end

% Momentum flux calculation using discrete integration (trapezoidal method)
xd7_momentum_flux = 0; % Initialize momentum flux
for i = 1:length(xd7_velocity)-1
    % Average radius and velocity squared between points for trapezoidal integration
    r_avg = (xd7_radius(i) + xd7_radius(i+1)) / 2; % Average radius between points
    v_sq_avg = ((xd7_velocity(i)^2) + (xd7_velocity(i+1)^2)) / 2; % Average of velocity squared
    dA = 2 * pi * r_avg * (xd7_radius(i) - xd7_radius(i+1)); % Annular area segment

    % Incremental momentum flux for the segment
    xd7_momentum_flux = xd7_momentum_flux + rho * v_sq_avg * dA;
end

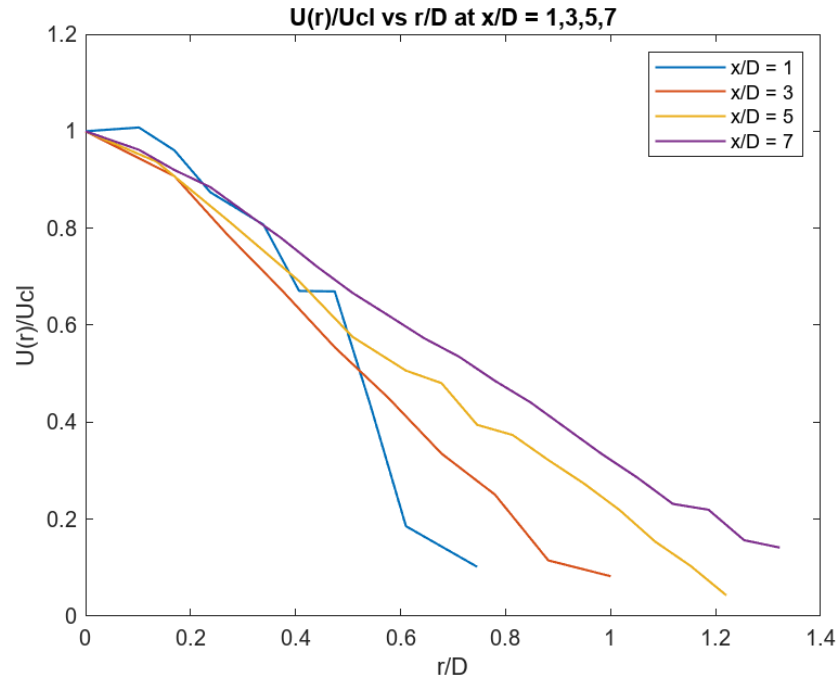
jet_momentum = [xd1_momentum_flux xd3_momentum_flux xd5_momentum_flux xd7_momentum_flux];

figure(13)
plot(xd, jet_momentum, 'o', 'LineWidth', 1)
title('Jet Momentum vs x/D')
xlabel('x/D')
ylabel('Jet Momentum (kg*m/s^2)')
xlim([0 8])
ylim([0 1])
grid on

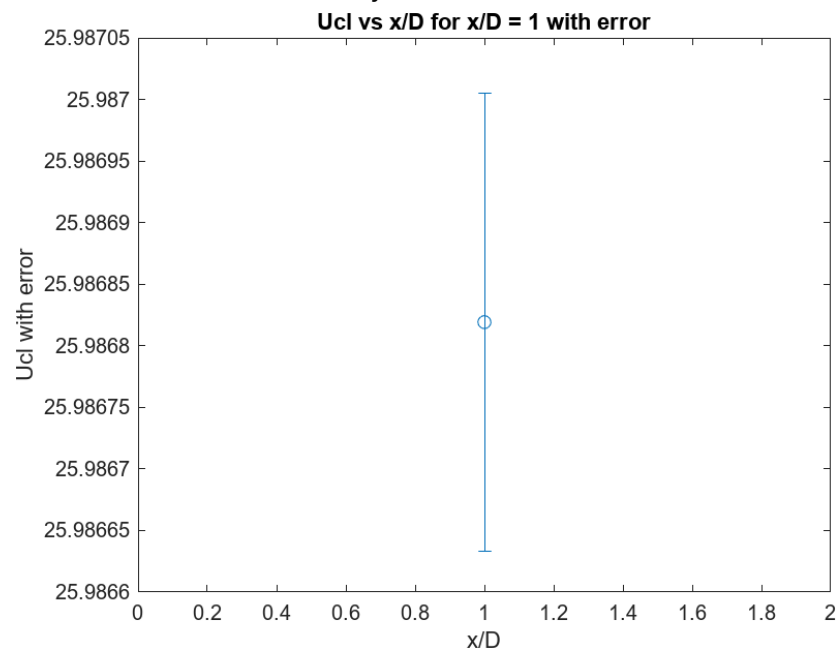
```

- (1) MATLAB Script was used for all data analysis. The numbering of sections relates to the order of topics in the results section. All output plots are in Appendix D. Raw data is available upon request.

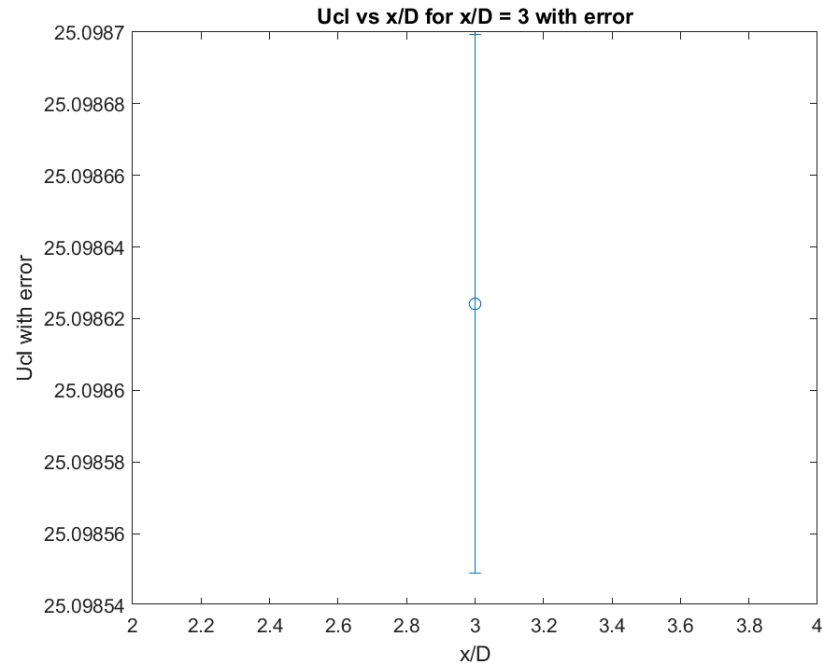
Appendix D - MATLAB Figures



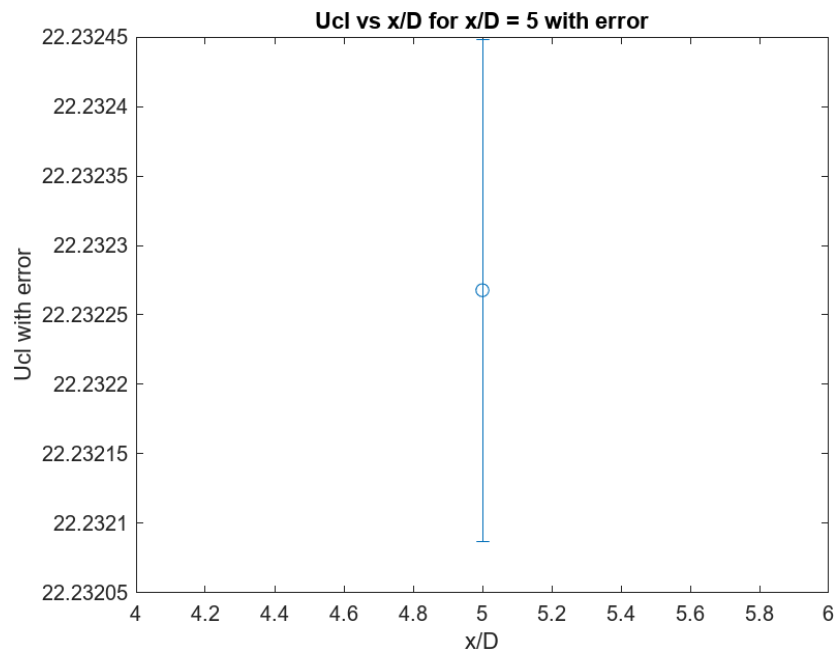
- (1) Normalization of velocity vs flow radius to nozzle diameter ratio



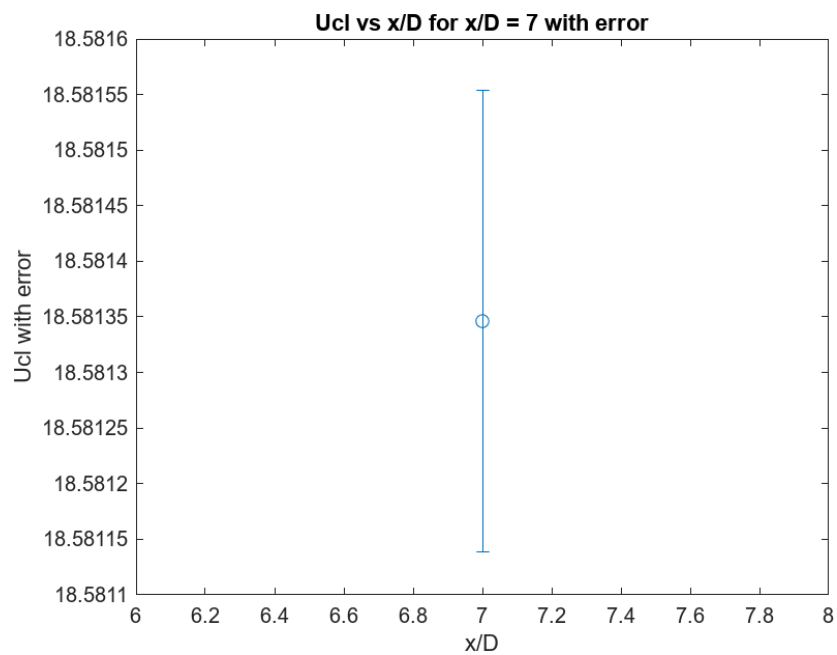
- (2) Centerline velocity vs $\frac{x}{D} = 1$ with error



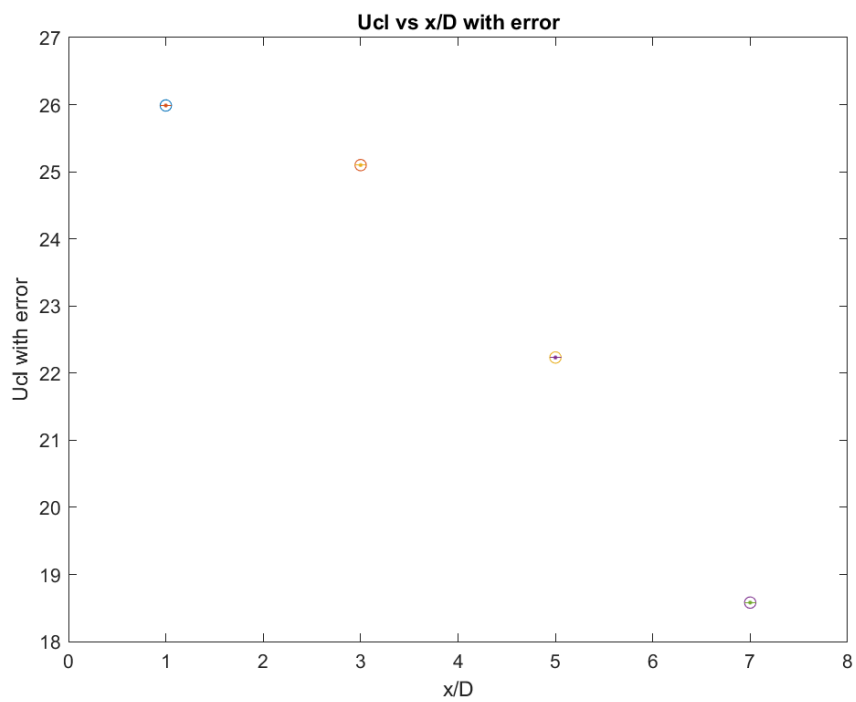
(3) Centerline velocity vs $\frac{x}{D} = 3$ with error



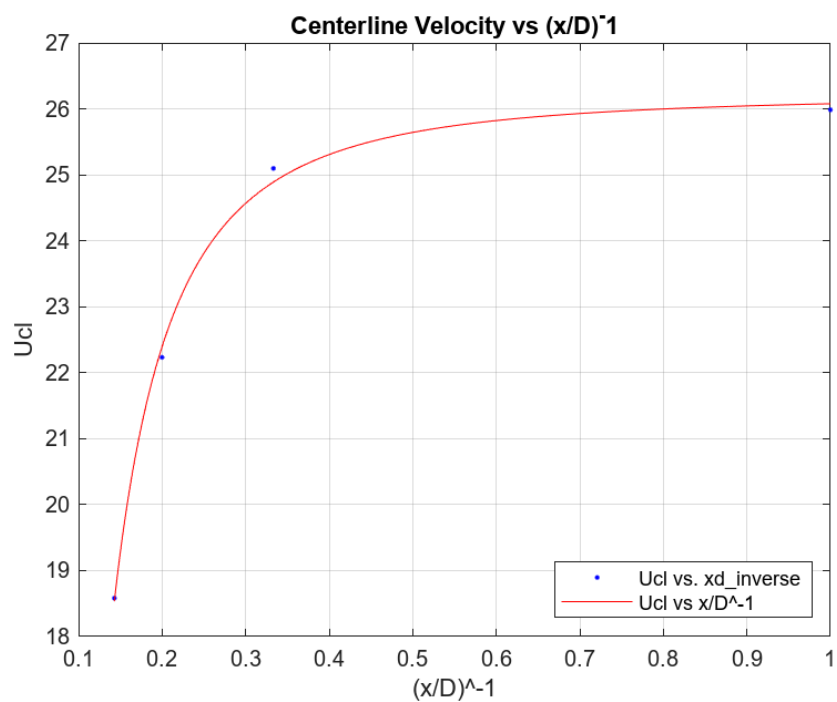
(4) Centerline velocity vs $\frac{x}{D} = 5$ with error



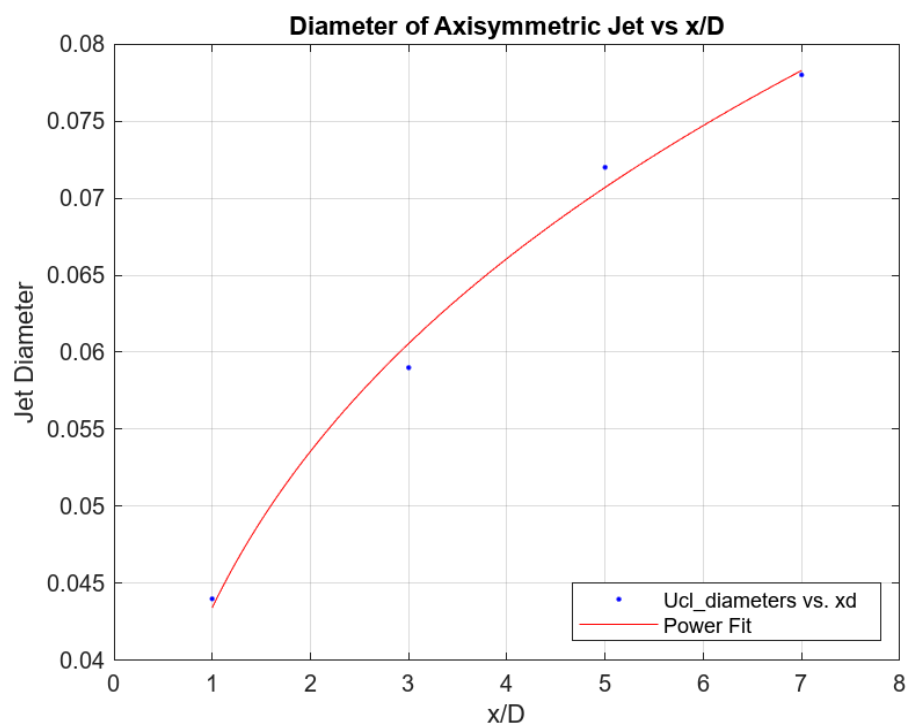
(5) Centerline velocity vs $\frac{x}{D} = 7$ with error



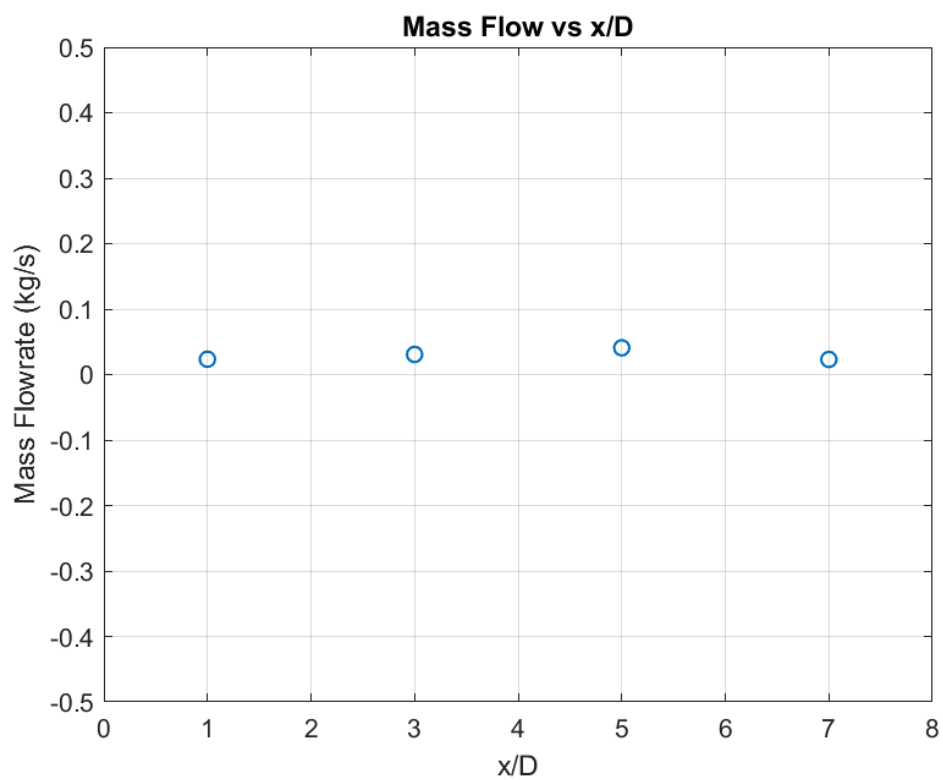
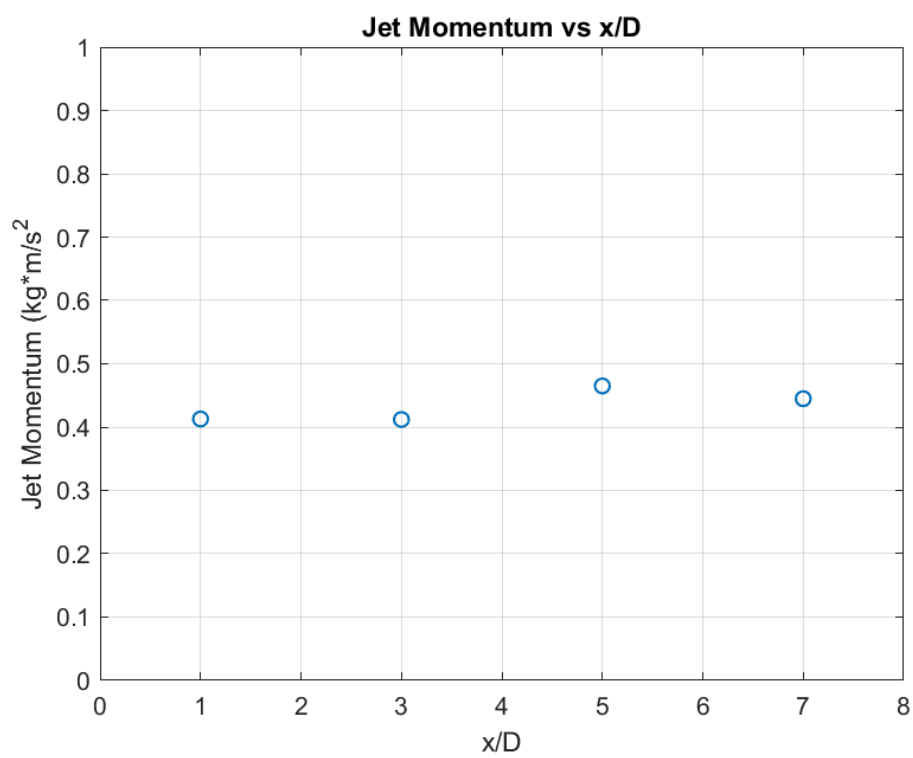
(6) Centerline velocity vs $\frac{x}{D}$ with error bars



(7) Centerline velocity vs $\frac{x}{D}^{-1}$



(8) Diameter of the Jet vs $\frac{x}{D}$

(9) Mass Flow vs $\frac{x}{D}$ (10) Jet Momentum vs $\frac{x}{D}$